Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

Author: Nikhil Vyas, Depen Morwani, Rosie Zhao, Gal Kaplun, Sham Kakade, Boaz Barak

Date: 2024



Motivation: Online Learning Context

- Online learning $singleepoch$ dominates LLM training $e.g., GPT-2, Llama$
- Prior work assumes SGD noise provides implicit bias advantages universally
- Critical question: Does SGD noise matter in online learning?



Golden Path Hypothesis

- Proposed: SGD path ≈ noiseless GD path in online learning
- Contrast with "Fork in the Road" $offline: noise changes final model$
- Figure 2: Illustrates both scenarios with key implications

Offline vs Online Learning Contrast

- **Offline learning** $multi-epoch$:
  - Smaller batch sizes → better minima $flatter, generalizebetter$
  - Figure 1 $left$: Lower batch size plateaus at lower loss
- **Online learning** $singleepoch$:
  - Larger batch sizes → better performance $noimplicitbias$
  - Figure 1 $right$: Higher batch size reaches lower loss

Theoretical Support $ConvexCase$

- Theorem 2.1: Quadratic loss with additive Gaussian noise
- $\mathbb{E}[\mathcal{L}(w_{R_1}(t))] \geq \mathbb{E}[\mathcal{L}(w_{R_2}(t))]$
- $\mathbb{E}[w_{R_1}(t)] = \mathbb{E}[w_{R_2}(t)] = w_{GD}(t)$
- SGD noise adds variance, not bias

Snapping Back to Golden Path

- Experiment: Reduce noise at T_0 steps $increasebatchsize$
- Result $Figure4$:
  - Loss sharply improves → follows low-noise trajectory
  - "Snaps" to golden path within $\tau \ll T_0$ steps
- Contrasts with offline learning $nosnappingatconvergence$

Function Space Evidence

- Measured total variation TV distance between models
- Golden path prediction:
  - After noise reduction, TV ≈ two low-noise models
  - Not ≈ high-noise vs low-noise models
- Figure 6: Confirms TV distance matches baseline

Implications & Conclusion

- **Practitioners** : Batch size choice driven by computation, not implicit bias
- **Theorists** : Focus noise effects on offline learning regime
- Takeaway: SGD in online learning = noisy GD steps on "golden path"

